Problem Statement

Elderly people suffering from dementia, Alzheimer’s disease, memory loss, or age-related cognitive disorders often forget routes and wander away from their homes or caregivers. In crowded public places such as streets, bus stands, railway stations, hospitals, and markets, locating missing elderly individuals becomes difficult and time-consuming. Existing solutions mainly rely on manual searching, isolated CCTV monitoring, or wearable GPS devices, which may fail if the device is removed, damaged, or unavailable.

There is currently no intelligent unified system that combines image analysis, video surveillance, audio assistance, and public reporting to support real-time tracking and prediction of missing elderly persons. Delays in locating them can expose elderly individuals to accidents, health risks, emotional stress, and unsafe environments.

Therefore, there is a need for a multimodal AI-based elderly tracking and assistance system that can intelligently identify, track, and predict the movement of missing elderly persons using connected camera feeds, uploaded images, audio samples, and citizen reports in order to assist families and authorities in locating them quickly and safely.